

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0961

Benton 138 kV
May 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0961. The Interconnection Customer proposes a 185.8 MW solar photovoltaic facility interconnecting at the Benton 138 kV substation in the New Jersey area. The request was received on March 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,673,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GI Dairy Biogas to Hills Chapel 138kV	Suffolkshire to United Power 138kV L	130.6	198	68.0
Alta to West Dallas Tap 138kV line	Hills Chapel to Burns 345kV	128.1	238	33.8
Livingston Westside to Fern Hill 138kV	Burns to Hills Chapel 345kV	122.6	261	33.8
Livingston Westside to United Power 138kV	Burns Substation to Hills Chapel 345kV	120.7	270	33.8
Benton to GI Dairy Biogas 138kV Line	Suffolkshire to United Power 138kV	116.3	195	86.7
Hills Chapel 161-138 kV TX	Livingston Westside to United Power 138kV	114.0	164	64.2
Hills Chapel 345 kV/161 kV XFMR	Wallenpaupack 500/345kV Auto	105.7	311	33.8
West Dallas Tap to Fern Hill 138kV	Burns to Hills Chapel 345kV	104.6	295	33.8
United Power to Bettas Rd 138kV	Burns to Hills Chapel 345kV	101.4	190	11.4
Burns 345/138kV XFMR	Burns to Coronado 345kV	101.1	478	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GI Dairy Biogas to Hills Chapel 138kV L	Rebuild line with higher-capacity conductor	\$17,499,000	100% (\$17,499,000)
Alta to W Dalas Tap 138kV line	Rebuild line with higher-capacity conductor	\$13,138,000	100% (\$13,138,000)
Fern Hill to Livingston Westside 138kV line	Rebuild line with higher-capacity conductor	\$6,602,000	100% (\$6,602,000)
Livingston Westside to United Power 138kV	Rebuild line with higher-capacity conductor	\$5,493,000	100% (\$5,493,000)
Benton to GI Dairy Biogas 138kV	Rebuild line with higher-capacity conductor	\$20,117,000	100% (\$20,117,000)
Hills Chapel 161-138 kV TX	Replace transformer with higher-rated unit	\$3,940,000	100% (\$3,940,000)
Hills Chapel 345 kV/161 kV XFMR	Replace transformer with higher-rated unit	\$21,950,000	79% (\$17,329,000)
West Dallas Tap to Fern Hill 138kV	Reconductor line	\$1,204,000	100% (\$1,204,000)
United Power to Bettas Rd 138kV	Reconductor line	\$2,693,000	100% (\$2,693,000)
Burns 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$13,658,000	100% (\$13,658,000)
	Total allocated to project		\$101,673,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.